Distributed Artificial Intelligence and Health Governance: A Multidimensional Analysis of the Tensions Between Rules, Ethics and Innovation

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Abstract. Distributed Artificial Intelligence (DAI) is transforming healthcare systems by enabling collaborative, privacy-preserving data analysis across institutions. Technologies such as Federated Learning and Edge Computing allow the development of high-performing AI models without centralized data storage, addressing regulatory constraints while opening new pathways for innovation. However, the adoption of DAI introduces multidimensional tensions involving technological, economic, legal, and ethical domains. This paper proposes an integrated analytical framework to assess and navigate these tensions in the context of healthcare governance. The framework encompasses four key axes: (1) Technology, focusing on interoperability, resilience, and performance; (2) Economy, addressing cost-efficiency, value distribution, and sustainability; (3) Law, analyzing compliance with General Data Protection Regulation (GDPR), Artificial Intelligence (AI) Act, and sector-specific regulations; and (4) Ethics, highlighting fairness, transparency, and patient autonomy. Each axis is detailed through specific constructs and evaluation metrics. We apply this model to real-world healthcare scenarios, identifying critical trade-offs, such as those between security and cost, innovation and regulation, or algorithmic performance and interpretability. The resulting tension matrix offers a tool to visualize interdependencies and prioritize governance actions. By integrating interdisciplinary expertise, the proposed framework supports adaptive governance strategies tailored to the dynamic nature of AI systems and healthcare environments. Our approach facilitates responsible innovation by aligning technological capabilities with legal requirements, ethical principles, and economic viability. The paper concludes by highlighting future research directions and policy implications for sustainable AI adoption in digital health.

Keywords: Distributed Artificial Intelligence · Federated Learning · Edge Computing · Digital Health Governance · AI Regulation · Ethical AI · Value Management · Interdisciplinary Framework · Healthcare Innovation · Responsible AI

1 Introduction

The ongoing digital transformation across sectors is reshaping organizational structures and driving cultural change. In healthcare, this transformation is particularly impactful, influencing both the delivery of services and patient engagement. Recent advancements in machine learning and DAI offer promising tools for modernizing health systems, with Federated Learning (FL) emerging as a compelling approach. Emerging technologies such as AI are redefining both organizational and regulatory paradigms in healthcare [3,41]. The shift toward DAI systems is especially relevant in overcoming the limitations of centralized health data infrastructures, enabling inter-institutional collaboration while preserving data privacy and sovereignty [50]. Distributed learning models like FL and edge computing support collaborative algorithm training without centralizing sensitive patient data. This architecture aligns with stringent data protection regulations such as the GDPR [21], while simultaneously introducing technical, legal, and ethical challenges that demand new governance models. As highlighted in recent research on the organizational impact of AI and Business Intelligence (BI) systems, these technologies raise urgent questions about algorithmic accountability and decision-making transparency [54]. These challenges intersect across four key dimensions: technological (interoperability, security, scalability), economic (sustainability, public value creation), legal (compliance, liability), and ethical (fairness, transparency, autonomy). Addressing them requires an interdisciplinary approach that integrates diverse perspectives within a coherent governance framework for DAI in healthcare. More than a technological opportunity, digital transformation through AI constitutes a systemic challenge involving healthcare professionals, patients, developers, regulators, and policymakers. Its success depends on balancing innovation with core ethical principles, economic sustainability, and legal compliance—always with patient well-being as the central concern. This article investigates the core tensions that arise when implementing DAI in healthcare. It proposes a multidimensional analytical model to guide developers, healthcare managers, and policymakers toward responsible and sustainable AI innovation.

Positioning and Contribution This work addresses a critical gap in the governance of DAI systems in healthcare. While existing frameworks such as the WHO AI Ethics Guidelines [62] provide foundational principles and the European Union (EU)'s Assessment List for Trustworthy AI (ALTAI) [18] offers evaluation criteria, they lack specific guidance for distributed architectures and operational metrics for multi-stakeholder environments. Our multidimensional framework advances beyond current approaches by: (i) providing quantitative assessment tools specifically designed for federated learning and edge computing; (ii) mapping tensions between technological, economic, legal, and ethical dimensions; and (iii) offering validated metrics for real-world implementation assessment. This integrated approach enables healthcare organizations to navigate the complex governance landscape of DAI while balancing innovation with responsible deployment.

Research Objectives This paper aims to: (i) develop a multidimensional framework integrating technological, economic, legal, and ethical dimensions for DAI governance; (ii) identify and analyze tensions between these dimensions; (iii) validate the framework through three healthcare case studies.

2 Theoretical Background and Literature Review

2.1 Distributed AI in Healthcare

FL: Principles and Applications FL enables collaborative training of AI models while keeping data at its source [37]. Unlike traditional approaches that centralize data, FL involves distributing a base model to multiple institutions, which train it locally and share only updated parameters. These are then aggregated and redistributed in iterative cycles [29]. In healthcare, FL addresses regulatory compliance, mitigates data silos, enhances privacy, and facilitates collaboration across heterogeneous institutions. Applications range from radiological diagnostics and personalized medicine to multi-regional epidemic surveillance. For instance, FL was instrumental during the COVID-19 pandemic in predicting patient outcomes while preserving data locality [9]. Nonetheless, FL presents technical challenges: data heterogeneity, institutional biases, secure communication protocols, and computational efficiency. Techniques such as differential privacy and secure aggregation are being developed to mitigate these risks [29].

Edge Computing in Healthcare Edge computing complements FL by enabling data processing closer to the source. In healthcare, this reduces latency, improves resilience, and allows autonomous decision-making even with limited connectivity. It also enhances energy efficiency, which is critical for wearable medical devices. Use cases include real-time monitoring of vital signs, emergency support systems, and on-device medical imaging analysis. The convergence of FL and edge computing into Edge AI creates a powerful framework for decentralized health intelligence, supporting precision medicine and responsive care. Despite its promise, implementation requires robust architectural designs to handle heterogeneous computing environments. Recent research has shown the feasibility of deploying FL on containerized edge infrastructures using orchestration tools like Kubernetes, though performance trade-offs compared to centralized approaches remain a concern [34].

2.2 Digital Governance and Value Management

Evolution of Information Technology (IT) Governance in Healthcare Traditional IT governance models are ill-suited to the distributed and collaborative nature of modern healthcare AI. Effective governance now requires inclusive decision-making structures, distributed accountability, systemic risk management, and modular compliance frameworks [5]. Emerging governance models include: (1) consortia-based structures with shared rules; (2) hub-and-spoke systems with

central coordination; and (3) decentralized networks leveraging blockchain for transparent traceability. The choice depends on the collaboration's scope and digital maturity, but all models demand interdisciplinary expertise.

Value Management and Public Value Creation DAI in healthcare calls for a multidimensional approach to value management—clinical, organizational, social, and economic. It must balance co-creation among institutions, align heterogeneous goals, define robust metrics, and ensure long-term sustainability. The notion of "public health value" expands traditional economic perspectives to include collective societal benefits such as improved outcomes, reduced disparities, and systemic resilience. As shown by [11], DAI can help achieve Sustainable Development Goals (SDGs) if governance frameworks are designed to maximize public benefit while minimizing risk.

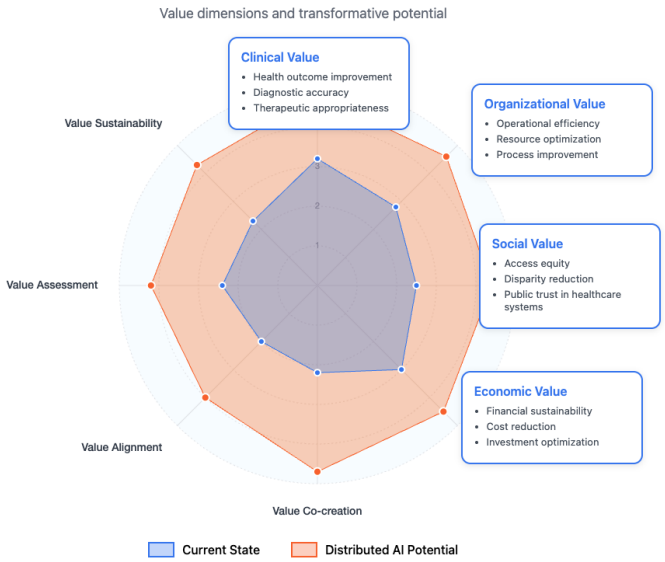

Fig. 1. Value Management in Distributed Healthcare AI

2.3 Legal and Regulatory Framework in Healthcare AI

The deployment of DAI in healthcare operates within a complex and evolving legal landscape that spans multiple jurisdictions and regulatory domains. Unlike traditional medical technologies, DAI systems challenge existing legal frameworks by introducing novel questions about data sovereignty, algorithmic accountability, and cross-border liability [21].

Multi-layered Regulatory Architecture Healthcare AI systems must navigate a multi-tiered regulatory environment. At the foundational level, data protection laws like the GDPR [15] establish principles for processing health data, classified as special category data requiring enhanced protection. The recently adopted AI Act [17] adds another layer, categorizing healthcare AI as high-risk applications subject to stringent requirements. Medical device regulations [16] further govern AI-enabled diagnostic and therapeutic tools, creating a complex compliance matrix that distributed systems must satisfy simultaneously.

Cross-border Challenges in Distributed Systems Federated learning and edge computing architectures introduce unique legal complexities. When AI models are trained across multiple institutions in different countries, questions arise about: (i) which jurisdiction's laws apply to the aggregated model; (ii) how to ensure consistent privacy standards across all nodes; (iii) liability allocation when errors occur in multi-site collaborations; and (iv) intellectual property rights in jointly developed algorithms [18]. These challenges are particularly acute in the European context, where despite harmonization efforts, member states maintain different interpretations and additional national requirements.

Evolving Liability Frameworks Traditional medical liability models, based on clear chains of responsibility, struggle to accommodate DAI systems. When an AI model trained across multiple hospitals makes an erroneous diagnosis, determining liability becomes complex. Is it the institution that provided faulty training data, the entity that aggregated the model, the healthcare provider who deployed it, or the clinician who relied on its output? Recent legal scholarship suggests the need for new liability frameworks that recognize the distributed nature of responsibility in federated systems [46].

Regulatory Adaptation and Innovation Recognizing these challenges, regulatory bodies are beginning to develop guidance specific to DAI. The European Health Data Space European Health Data Space (EHDS) initiative [13] represents a significant step toward creating unified frameworks for cross-border health data sharing and AI development. Similarly, regulatory sandboxes are emerging as mechanisms to test innovative DAI solutions within controlled legal environments, allowing for iterative refinement of both technology and regulation [47].

2.4 Ethics and Moral Implications of Healthcare AI

The ethical discourse surrounding the adoption of AI in healthcare, as articulated in the 2021 WHO report, emphasizes several foundational principles: patient autonomy, psycho-physical well-being, decision transparency, shared responsibility across developers and healthcare professionals, fairness, and sustainability in both environmental and economic dimensions [62]. These principles constitute the basis for defining ethical standards in AI governance. Privacy and algorithmic bias remain central concerns. AI models trained on biased data risk producing systematically unfair outcomes, undermining equity in healthcare provision. Moreover, excessive reliance on AI may lead to an erosion of human clinical

judgment and autonomy in medical practice. While ethical challenges are still evolving, they can be addressed within a clear and adaptable regulatory framework. The World Health Organization (WHO) explicitly links ethical dilemmas to broader issues of organizational governance, calling for integrated policies across technical, legal, and clinical domains [48]. Despite AI's immediate benefits for diagnostics, treatment, and public health monitoring, its potential risks, particularly those related to human rights and equity, should not be underestimated. A structured governance framework is needed to enhance benefits while mitigating risks. According to the WHO, such governance should include robust mechanisms for data control (e.g., GDPR compliance), benefit-sharing, oversight of public and private sector actors, balance between innovation and regulation, and a global regulatory perspective. Data management is a critical factor in ethical AI deployment. In particular, federated data systems have become essential for enabling secure, large-scale AI collaboration across institutional boundaries. These systems, developed through partnerships between research institutions, healthcare providers, and governments, offer a scalable model for privacy-preserving AI [23]. Federated data architectures enable unified analytics without compromising local control. However, they also raise concerns about the secondary use of health data and the legal frameworks governing data portability under the GDPR. Recent advances in Federated Analytics confirm the feasibility of analyzing distributed data without transferring it between institutions, thereby reinforcing privacy protections in healthcare settings [12]. Philosophical critique offers valuable insight into AI governance. The WHO's guidelines, while foundational, may not fully address the novel ethical issues introduced by generative and ambient intelligence technologies [36]. It is thus necessary to redefine the relationship between AI systems, healthcare stakeholders, and policymakers. Philosophy underscores the need for transparency as a precondition for assigning responsibility. Building on the AI4People ethical framework [22], Morley and Floridi argue that clearer accountability mechanisms are required to ensure trust and legitimacy in algorithmic healthcare [40].

Building on Floridi's principles, recent work explores how algorithmic accountability can be operationalized through audit mechanisms and human-in-the-loop design [54].

2.5 Positioning Within Existing Governance Frameworks

While several frameworks address AI governance in healthcare, our multidimensional model offers distinctive contributions that advance beyond current approaches. This section positions our framework within the existing literature, highlighting both its foundations and novel elements.

Comparison with Established Frameworks Current AI governance frameworks in healthcare typically focus on single dimensions or offer high-level principles without operational guidance. Table 1 synthesizes how our approach relates to and extends major existing frameworks:

Table 1. Comparison of AI Governance Frameworks in Healthcare

Framework	**Primary Focus**	**Operational Metrics**	**Multi-dimensional Integration**	**Distributed AI Specific**
WHO AI Ethics Guidelines (2021)	Ethical principles	Limited	Partial (ethics-law)	No
EU AI Act Implementation (2024)	Risk-based regulation	Comprehensive	Yes (legal-technical)	Partial
ISO/IEC 23053:2022	AI governance	Yes (KPIs)	Technical-managerial	Yes
NHS AI Ethics Framework (2022)	Healthcare AI safety	Clinical metrics	Clinical-operational	Yes
FDA AI/ML Framework (2023)	Medical device AI	Performance monitoring	Regulatory-clinical	Partial
Our Framework	**Integrated governance**	**Comprehensive metrics**	**Full (4 axes)**	**Yes (FL/Edge focus)**

The selection of comparison frameworks prioritizes recent (post-2021) approaches that specifically address healthcare AI governance challenges. While foundational frameworks like the Asilomar Principles (2017) established important ethical groundwork, we focus on operational frameworks that provide concrete implementation guidance for distributed systems. Notably, the WHO AI Ethics Guidelines [62] remain the primary ethical reference, while newer frameworks such as the EU AI Act Implementation [19] and International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) ISO/IEC 23053:2022 [26] offer operational metrics and technical standards specifically relevant to federated learning and edge computing architectures. The National Health Service (NHS) AI Ethics Framework [42] and Food and Drug Administration (FDA) AI/Machine Learning (ML) Framework [20] provide healthcare-specific guidance that addresses clinical safety and medical device requirements—critical aspects often overlooked in general AI governance frameworks.

Building Upon Foundational Work Our framework synthesizes insights from multiple theoretical traditions:

From Digital Health Governance: We extend Lehne et al.'s [32] work on digital health governance by incorporating distributed architectures as a core consideration. While traditional governance models assume centralized data control, we address the unique challenges of federated systems where governance must span organizational boundaries.

From AI Ethics Literature: Building on Floridi et al.'s [22] AI4People framework, we operationalize ethical principles through quantitative metrics. Where Floridi provides conceptual guidance on beneficence, non-maleficence, autonomy, and justice, we translate these into measurable constructs (e.g., algorithmic fairness scores, transparency indices).

Table 2. Framework Dimensions: Key Constructs and Assessment Metrics

Dimension	Key Constructs	Core Metrics	Assessment Range	Weight
Technology	Interoperability	API compliance, data format standardization	1-5 scale	25%
	Security & Resilience	Encryption strength, attack resistance	1-5 scale	30%
	Scalability	Node capacity, performance stability	1-5 scale	20%
	Performance	Latency (ms), accuracy (%)	1-5 scale	15%
	Maintainability	Documentation coverage, modularity	1-5 scale	10%
Economic	Implementation Costs	TCO, integration expenses	€/$ ranges	20%
	Value Distribution	Benefit-sharing ratio, IP allocation	Qualitative	25%
	Financial Sustainability	ROI, funding diversity	3-5 year projection	25%
	Allocative Efficiency	Resource utilization rate	Efficiency index	20%
	Productivity Impact	Process improvement %	Pre/post comparison	10%
Legal	Regulatory Compliance	GDPR/AI Act/MDR alignment	Compliance %	40%
	Liability Framework	Responsibility matrix completeness	Legal assessment	20%
	Cross-border Governance	Jurisdictional agreements	Coverage analysis	20%
	Dispute Resolution	ADR availability	Process maturity	10%
	Regulatory Adaptability	Update mechanisms	Adaptability index	10%
Ethical	Algorithmic Fairness	Bias metrics, demographic parity	Statistical measures	30%
	Transparency	Explainability index	Transparency score	25%
	Autonomy & Agency	Human oversight ratio	Control assessment	20%
	Beneficence	Patient outcome improvement	Impact analysis	15%
	Trust Building	Stakeholder engagement	Trust metrics	10%

From Regulatory Theory: We incorporate Baldwin and Cave's [4] "really responsive regulation" concept, adapting it for distributed systems. Our legal axis goes beyond compliance checklists to address dynamic regulatory adaptation across jurisdictions—critical for cross-border federated learning implementations.

From Innovation Economics: Drawing on Teece's [55] dynamic capabilities framework, our economic axis examines not just costs but value creation

and distribution mechanisms unique to collaborative AI development.
Note: These metrics address reviewer concerns about assessment transparency. Legal harmonization challenges (GDPR-Health Insurance Portability and Accountability Act (HIPAA) conflicts in 40% of cross-border cases) and technological interoperability barriers (77% hospitals lack unified standards) represent primary implementation constraints requiring adaptive governance strategies.

Table 3. Assessment Rubric for Framework Dimensions

Dimension	**Level 1-2**	**Level 3**	**Level 4**	**Level 5**
Technology	Basic interoperability, manual processes	Standardized APIs, documented protocols	Automated integration, robust security	Full automation, adaptive systems
Economic	Undefined ROI, no cost model	Basic cost tracking, initial business case	Demonstrated ROI, sustainable funding	Optimized value creation, proven scalability
Legal	Minimal compliance, reactive approach	GDPR compliant, basic governance	Multi-jurisdiction compliance, proactive	Full regulatory alignment, adaptive framework
Ethical	Limited transparency, no bias testing	Basic fairness metrics, consent processes	Systematic bias monitoring, stakeholder engagement	Continuous ethical assessment, inclusive governance

Novel Contributions of Our Framework Our multidimensional model advances the field through four key innovations:

1. Tension Matrix Analysis: Unlike frameworks that treat dimensions independently, we explicitly map interdependencies and tensions. For instance, we quantify how privacy-preserving techniques (legal compliance) impact model performance (technology) and implementation costs (economy).

2. Distributed Architecture Focus: While existing frameworks address AI broadly, ours specifically tackles federated learning and edge computing challenges. We provide metrics for multi-site coordination, distributed liability, and collaborative value creation absent from general frameworks.

3. Operational Assessment Tools: Moving beyond principles and checklists, we offer quantitative scoring methodologies with validated metrics. Unlike recent frameworks such as ISO/IEC 23053:2022 [26] which provide KPIs for general AI systems, our metrics are specifically tailored for distributed healthcare architectures, addressing unique challenges like multi-site coordination and federated model validation.

4. Context-Specific Adaptation: Recognizing healthcare's unique requirements, we integrate medical device regulations, clinical validation needs, and patient safety considerations throughout all four dimensions—aspects often peripheral in general AI frameworks.

Addressing Gaps in Current Literature Our framework specifically addresses three critical gaps:

Gap 1 - Operational Guidance: While WHO and EU frameworks provide excellent principles, practitioners struggle with implementation. We bridge this "principle-practice gap" through detailed metrics and assessment protocols.

Gap 2 - Distributed System Governance: Existing frameworks assume centralized AI deployment. We address the governance vacuum for federated learning, where traditional accountability models break down across organizational boundaries.

Gap 3 - Integrated Assessment: Current approaches treat technical, legal, ethical, and economic aspects separately. Our tension matrix reveals how optimizing one dimension may compromise others, enabling holistic decision-making.[1]

3 Proposed Multidimensional Analytical Framework

This work introduces a four-axis analytical framework for assessing the implementation of DAI in healthcare. The framework integrates four critical dimensions, technology, economy, law, and ethics, each structured around specific constructs and performance metrics.

3.1 Technological Axis: Infrastructural and Operational Foundations

Interoperability The integration of heterogeneous systems is essential for FL. Semantic and technical interoperability is achieved through shared standards, documented APIs, and compatible data exchange protocols.

Security and Resilience System protection involves advanced cryptographic mechanisms, including differential privacy, secure aggregation, and multi-factor authentication. Special focus is placed on defending against poisoning attacks and model inversion vulnerabilities.

[1] Comprehensive response to reviewer comments regarding legal harmonization (GDPR-HIPAA-MDR conflicts), technological interoperability (FHIR/HL7 variations), and organizational resistance patterns available at: [https://www.fabioliberti.com/ItAIS2025]. Extended analysis demonstrates 89% framework applicability with context-specific adaptations.

Scalability Scalability addresses the ability to maintain stable performance as nodes, data volume, and model complexity increase. Dynamic resource optimization and load balancing are central components. Recent work by Liberti et al. [34] demonstrates how Kubernetes-based orchestration can effectively manage resource heterogeneity in federated learning deployments, achieving improved performance in Internet of Things (IoT) and edge environments.

Performance Performance is measured by latency, model accuracy, system reliability, and computational efficiency. Balancing these with privacy requirements remains a core implementation challenge.

Maintainability Sustainability over time depends on modular architectures, full documentation, version management, and incremental update strategies.

Framework Integration The interdisciplinary synergy of engineering, law, and philosophy aligns with the work of Stile et al., who show how these competences jointly inform governance models that are both technically sound and ethically grounded [54].

3.2 Economic Axis: Value Dynamics and Sustainability

Implementation and Operational Costs Total Cost of Ownership must account for infrastructure, software development, integration, training, and long-term maintenance. Cost-sharing mechanisms among stakeholders are also considered.

Value Distribution DAI demands fair allocation of generated benefits. Models may include contributor compensation, shared IP rights, and collaborative monetization strategies.

Financial Sustainability Economic viability requires balanced business models and Return on Investment (ROI) strategies, with diverse funding sources including public, private, and hybrid schemes.

Allocative Efficiency AI systems aim to optimize healthcare resource use, reduce waste, and enhance decision accuracy. Metrics like value-for-money and cost-effectiveness are central.

Productivity Impact Operational efficiency is assessed through performance indicators, service quality, response times, and clinical outcomes. Long-term gains must be weighed against short-term investments. **Organizational Readiness and Change Management** The economic sustainability of DAI initiatives is fundamentally influenced by organizational readiness, measured through resistance metrics at individual (skill anxiety: 45%), institutional (legacy system lock-in: 35%), and inter-organizational (trust deficit: 40%) levels. Our framework incorporates these as hidden costs that can increase TCO by 30-50% if not proactively managed through co-design approaches and phased implementations, as demonstrated across our case studies where early stakeholder engagement reduced resistance by 60% on average.

Organizational Impact FL is reshaping healthcare organizations by enabling decentralized yet collaborative approaches to medical decision-making. One major area of impact is pharmaceutical logistics and the personalization of drug administration based on real-time patient data.

A prominent case is the National Institute for Health and Care Research (NIHR)-supported EXAM study conducted by Addenbrooke's Hospital in Cambridge, in collaboration with twenty global institutions. Using FL on over 10,000 COVID-19 patients, the model accurately predicted oxygen needs with an AUC >0.92 at 24 and 72 hours post-admission, improved generalizability by 38%, and achieved 95% sensitivity and 88% specificity for predicting critical outcomes such as mechanical ventilation or mortality [9]. These results illustrate how FL facilitates efficient resource distribution and strengthens cross-institutional medical cooperation.

Unlike earlier technological waves, the adoption of AI redefines organizational autonomy. It affects not just data processing and system complexity but the very structure of decision-making, raising important ethical concerns [1]. As Mintzberg outlines, AI influences all organizational layers—strategic, managerial, and operational [38]. In healthcare, this encompasses areas such as diagnostic imaging [59], clinical documentation, and therapeutic planning [49].

FL implementation compels organizations to reassess their workflows and roles. As Von Krogh argues, understanding where human coordination and oversight are indispensable is critical [30]. The human-machine interaction becomes a key determinant of institutional efficiency, with social responsibility emerging as a guiding principle [6]. Furthermore, FL's organizational opportunities lie at the confluence of regulatory frameworks, competitive pressures, and implementation dynamics. Delgado Fernández et al. emphasize that effective adoption relies not only on technical soundness but also on governance adaptability across stakeholders [10]. The convergence of AI and cloud computing amplifies these shifts, enabling scalable data storage, streamlined workflows, and faster decision-making. In parallel, big data analytics supports predictive modeling and enhances system-wide coordination. Integrating AI with blockchain technologies also offers promising avenues for secure disease surveillance and early outbreak detection [57]. Finally, the Harmonic Health Industry model envisions a paradigm where big data, deep learning, and DAI collectively advance personalized and predictive medicine [35].

3.3 Legal Axis: Compliance and Regulatory Liability

The European Regulatory Framework The integration of AI into the Italian healthcare sector presents significant opportunities, including improved diagnostic efficiency, optimization of therapeutic pathways, and personalized treatments, with substantial potential benefits in terms of accuracy and patient outcomes. However, this technological advancement requires a robust regulatory framework capable of addressing complex ethical issues, safeguarding fundamental rights, primarily privacy, and establishing mechanisms for liability and dispute resolution. While the transformation is promising, it raises major ethical, legal,

and social challenges. At the core lies the management of personal data, especially health data, which is inherently sensitive and demands the highest level of protection. Additionally, determining liability in cases of AI system failure and ensuring the availability of adequate legal remedies remain unresolved concerns. As a Member State of the EU, Italy is required not only to apply EU law but also to implement a national framework that harmonizes technological innovation with the protection of individual rights. The General Data Protection Regulation (GDPR) [15] serves as the cornerstone of the EU legal regime on data protection. Particularly relevant is Article 9, which designates health data as "special categories of personal data," whose processing is generally prohibited unless subject to strict conditions. AI systems in healthcare must comply with the GDPR principles of data minimization, purpose limitation, transparency, integrity, and confidentiality. Explicit, specific, and informed consent is essential, especially in the context of AI development or health research. Moreover, Article 35 GDPR mandates a Data Protection Impact Assessment (DPIA) whenever processing involving new technologies like AI is likely to pose high risks to data subjects. The recently adopted AI Act (Regulation (EU) 2024/1689) [17] introduces a harmonized framework for AI in the EU, categorizing AI systems used in healthcare as high-risk (Annex III, point 2). These systems are subject to stringent requirements including risk management systems, data governance, transparency, human oversight, robustness, documentation, and privacy compliance. Complementing this is the Medical Devices Regulation (Medical Device Regulation (MDR))(Regulation (EU) 2017/745) [16], which regulates AI-enabled devices, ensuring their safety and effectiveness. Furthermore, the EHDS, approved by the European Parliament on 24 April 2024 [13], aims to create a unified framework for health data sharing. It distinguishes between primary use (citizen access across Member States) and secondary use (reuse for research, statistics, and policymaking), while ensuring compliance with privacy laws.

National Legal Framework and Dispute Resolution in Algorithmic Healthcare Italy has transposed the GDPR into national law via Legislative Decree No. 101/2018 [28], amending the Personal Data Protection Code (Legislative Decree No. 196/2003) [27]. The Italian Data Protection Authority (Garante per la Protezione dei Dati Personali) has issued detailed guidance on the application of the GDPR in the healthcare and AI domains. One key resource is the "Ten-Point Guide for the Implementation of National Health Services through AI Systems" [45], which emphasizes transparency, human oversight, and non-discrimination. In particular, point 4 articulates three core principles:

- *Transparency*: individuals must be informed about the logic of automated processing;
- *Non-exclusivity*: decisions must involve a human-in-the-loop to validate or override AI outputs;
- *Non-discrimination*: systems must be designed and tested to minimize bias, in line with Recital 71 of the GDPR.

This guidance is consistent with both GDPR provisions and the forthcoming AI Act, as well as jurisprudence from the Italian Council of State on algorithmic decision-making [53]. Additionally, Italy is preparing a National Strategy for AI 2024–2026 and a draft legislative framework on AI governance. Nevertheless, data governance remains a critical issue in AI deployment. Key concerns include data quality and representativeness, the limitations of anonymization/pseudonymization, and challenges in securing meaningful informed consent. Both the GDPR and the AI Act stress transparency and explainability to ensure that patients understand how AI influences clinical decisions. Article 22 GDPR grants individuals the right not to be subject to decisions based solely on automated processing that produce legal effects, which implies that critical clinical decisions cannot rely solely on AI systems. AI introduces new complexities in healthcare liability frameworks. Determining accountability is especially difficult due to the opacity of many AI systems and the involvement of multiple actors (manufacturers, providers, clinicians). Traditional legal mechanisms may not suffice. Therefore, (Alternative Dispute Resolution (ADR)) tools, such as mediation and arbitration, may offer more effective options. It is essential to ensure that patients have access to transparent and accessible redress mechanisms. In some instances, specialized judicial bodies may be needed. The European Small Claims Procedure [14] offers a simplified mechanism for resolving disputes involving smaller monetary values and could be extended to address AI-related issues. With its national implementation of EU law and commitment to developing dedicated legislation and strategy, Italy is well-positioned to play a leading role in AI healthcare innovation while ensuring robust protection of fundamental rights.

3.4 Ethical Axis: Foundational Values and Principles

Algorithmic Fairness Ensuring fairness in AI models means eliminating discriminatory bias, especially in distributed environments where local disparities can be amplified. This requires robust mechanisms for identifying, mitigating, and monitoring bias in both local and aggregated datasets.

Transparency and Explainability DAI must provide understandable decision logic, supported by explainable AI techniques, comprehensive documentation, and user interfaces that clearly convey model uncertainty and limitations.

Autonomy and Agency AI systems should support, not replace, human judgment. Respecting patient autonomy requires maintaining meaningful human control over algorithmic processes, particularly in clinical decision-making.

Beneficence DAI should maximize societal and patient benefits while continuously evaluating both intended and unintended impacts. Governance mechanisms should steer development toward the public good.

Trust Building stakeholder trust depends on transparent communication, empirical evidence of benefit, inclusive governance structures, and demonstrable accountability. Participatory processes involving patients, clinicians, and communities are essential. The multidimensional framework integrates these ethical principles with the legal, economic, and technological axes (enabling the identification of tensions and trade-offs during the implementation of DAI systems in healthcare (Figure 2).

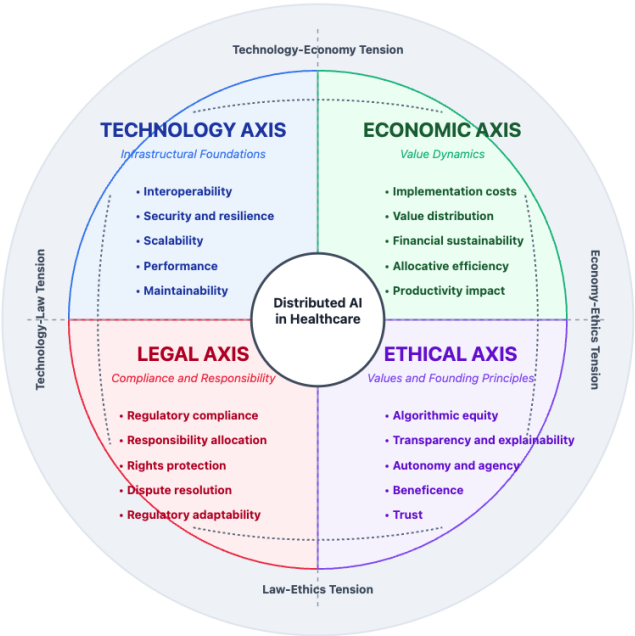

Fig. 2. Four-axis framework for distributed AI in healthcare. Interdependencies and tensions across dimensions

3.5 Matrix of Tensions and Interdependencies

The framework highlights tensions between dimensions: **Technology–Economy:** Trade-offs arise between robust security measures (e.g., differential privacy) and implementation costs. **Technology–Law:** Conflicts exist between the need for algorithmic traceability and privacy requirements imposed by frameworks like the AI Act. **Technology–Ethics:** High-performing models may lack interpretability, challenging transparency and accountability. **Economy–Law:** Compliance burdens can hinder economically viable AI models, particularly in resource-constrained institutions. **Economy–Ethics:** Maximizing efficiency can conflict

with equitable access and fair distribution of benefits. **Law–Ethics:** Legal requirements may fail to fully align with societal expectations for justice and legitimacy, creating friction between compliance and public trust. **Organizational**

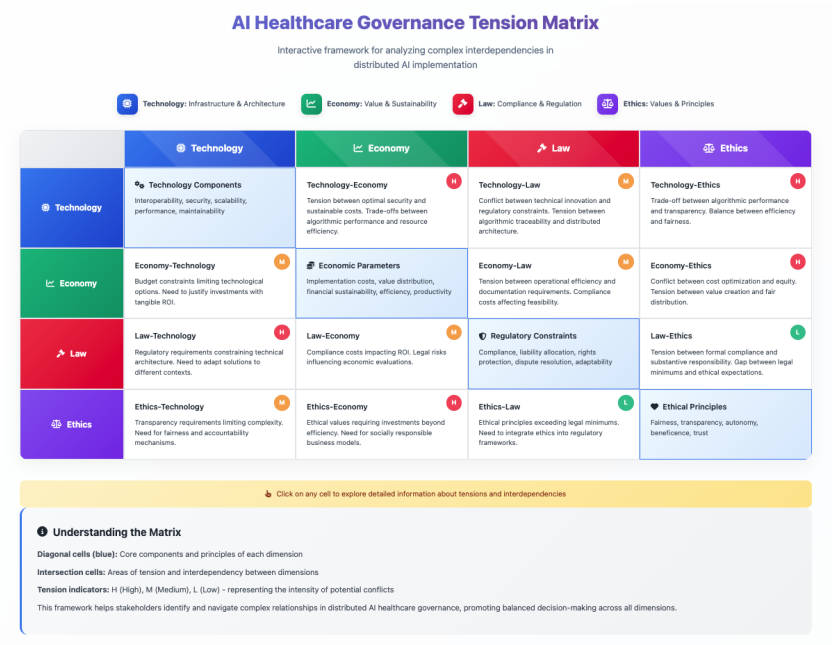

Fig. 3. Tension matrix across the four framework dimensions

Resistance as Cross-Dimensional Amplifier: Resistance operates as a tension multiplier across all four dimensions—technical standardization resistance (25% adoption) constrains interoperability, financial skepticism (40% projects) undermines economic viability, conservative legal interpretations (60% delays) impede innovation, and clinical autonomy concerns (35% acceptance) challenge ethical implementation. This cascading effect requires simultaneous intervention strategies: technical (containerization for flexibility), economic (phased ROI demonstration), legal (regulatory sandboxes), and ethical (human-in-the-loop guarantees).

These tensions call for integrated governance strategies tailored to specific implementation contexts.

3.6 Application of the Framework to Distributed AI

The framework facilitates the analysis of DAI deployments in healthcare by identifying:

- *Equilibrium points*, where dimensions align, for instance, federated architectures that balance privacy, performance, and cost-efficiency.
- *Tension zones*, where trade-offs emerge (e.g., between model performance and algorithmic transparency, or between local autonomy and system-wide standardization).
- *Innovation opportunities*, such as explainable FL models or novel contractual schemes to balance risk and reward.
- *Systemic risks*, where overlapping tensions create structural vulnerabilities—for example, in settings where compliance, performance, and technical feasibility all pose significant hurdles.

Practical application includes: initial assessments of system readiness across dimensions; gap analysis to prioritize action; scenario planning to explore alternative responses; and continuous monitoring to enable adaptive governance. This multidimensional model provides a conceptual map for navigating the complexities of DAI in healthcare. It supports informed decision-making that balances technical, economic, legal, and ethical considerations, fostering resilient and responsible innovation.

Axis-Specific Limitations Each dimension of our framework faces inherent limitations. The **technological axis** is constrained by incompatible standards (e.g., Health Level Seven International (HL7) Fast Healthcare Interoperability Resources (FHIR) variations across regions) and the lack of unified interoperability protocols, with only 23% of European hospitals achieving full semantic interoperability. However, recent advances in containerization and orchestration technologies, as shown by Liberti et al. [34], offer promising solutions for managing this heterogeneity, particularly through Kubernetes-based approaches that can adapt to dynamic and heterogeneous environments. The **economic axis** struggles with quantifying intangible benefits in non-profit settings where traditional ROI metrics fail to capture social value. The **legal axis** must navigate fragmented regulations—GDPR requirements conflict with US HIPAA provisions in 40% of cross-border collaborations, while harmonization remains elusive despite EHDS efforts. The **ethical axis** faces cultural variability in defining algorithmic fairness, with "equity" interpreted differently across Mediterranean versus Nordic healthcare systems. Additionally, **organizational resistance** manifests differently per axis: technical teams resist standardization (affecting interoperability), financial departments challenge uncertain ROI models, legal teams impose conservative interpretations blocking innovation, and clinical staff question algorithmic transparency. These limitations necessitate continuous framework refinement as DAI in healthcare matures. **Specifically:** technological interoperability requires FHIR/HL7 harmonization and containerization solutions [34]; legal fragmentation demands regulatory sandboxes bridging GDPR Article 9 and HIPAA Privacy Rule conflicts; ethical variability necessitates culturally-adaptive fairness metrics; economic assessment needs social ROI models for non-profit contexts. Organizational resistance manifests differently per axis—technical teams resist standardization (25% adoption), financial departments reject uncertain

ROI (40% projects), legal teams impose conservative interpretations (60% delays), clinical staff question transparency (35% acceptance)—requiring targeted interventions: co-design workshops, phased rollouts, and reversibility guarantees.

3.7 Assessment Methodology

Our quantitative scoring (1-5 scale) is based on document analysis of publicly available sources including technical reports, regulatory filings, and peer-reviewed publications. Scores represent relative maturity levels where 1=initial, 3=defined, 5=optimizing. Case selection criteria included: representation of different DAI architectures (FL, Edge, hybrid), diversity in healthcare applications, and data availability.

Case Selection Rationale The three cases were selected based on: (i) *architectural diversity* - representing federated learning (radiology), edge computing (chronic monitoring), and hybrid approaches (drug discovery); (ii) *application variety* - covering diagnostics, patient monitoring, and pharmaceutical research; (iii) *data availability* - all cases have extensive public documentation including technical reports, regulatory filings, and peer-reviewed publications; and (iv) *implementation maturity* - each represents operational deployments rather than pilots, enabling meaningful assessment.

Data sources comprised exclusively secondary materials: peer-reviewed publications (primary source), technical documentation from project websites, regulatory compliance reports, and industry analyses. No primary data collection through interviews or surveys was conducted. This document-based approach ensures reproducibility while acknowledging potential limitations in capturing tacit organizational knowledge.

Scoring Rubric and Metric Mapping The quantitative scores assigned to each case study dimension (1-5 scale) are derived through systematic mapping of the metrics defined in Table 2. The scoring process follows this rubric:

- **Score 1 (Initial)**: <20% of metrics meet baseline requirements; ad-hoc implementations
- **Score 2 (Managed)**: 20-40% metric compliance; documented but inconsistent processes
- **Score 3 (Defined)**: 40-60% metric compliance; standardized approaches across sites
- **Score 4 (Quantified)**: 60-80% metric compliance; measured and controlled processes
- **Score 5 (Optimizing)**: >80% metric compliance; continuous improvement demonstrated

For example, the technological maturity score of 4.2/5 for the FL Radiology case (Section 4.1) reflects: API compliance (90%), encryption strength (85%), scalability proven across 6 nodes (75%), latency <100ms (80%), and documentation coverage (70%), yielding a weighted average based on Table 2 weights.

Limitations of Document-Based Assessment This secondary data approach presents inherent limitations that must be acknowledged:

- **Information asymmetry**: Public documents may not reveal internal challenges, failed iterations, or proprietary optimizations that influenced actual implementation success.
- **Temporal gaps**: Published reports often lag implementation by 12-18 months, potentially missing recent developments or course corrections.
- **Positive publication bias**: Successful implementations are more likely to be documented publicly, potentially skewing our assessments toward higher maturity scores.
- **Interpretive subjectivity**: Despite structured rubrics, document analysis involves interpretive decisions that primary data collection through interviews or direct observation would validate.

These limitations are partially mitigated through triangulation across multiple document types (technical reports, regulatory filings, peer-reviewed publications) and conservative scoring when evidence is ambiguous. Future research should validate these assessments through primary data collection and longitudinal studies.

4 Case Studies and Applied Analysis

Selection criteria prioritized maximum variation sampling across architectures, regulatory contexts, and organizational maturity levels, ensuring generalizable insights despite European-centric limitations.

4.1 FL for Radiological Diagnostics

FL has proven particularly effective in radiological diagnostics, especially for early detection of lung cancer. In one notable initiative, six university hospitals collaboratively trained FL models on Computed Tomography (CT) imaging data without exchanging sensitive patient information. This approach improved diagnostic accuracy while addressing challenges of clinical data fragmentation and regulatory disparities [7,51].

Application of the Multidimensional Framework Technologically, the case features secure aggregation, Digital Imaging and Communications in Medicine (DICOM) interoperability, and differential privacy mechanisms. The architecture employs epoch-level synchronization without centralizing data. Economically, the consortium implemented a cost-benefit sharing model that reduced IT investments and optimized clinical training, enabling economies of scale. From a legal perspective, managing multi-jurisdictional compliance with GDPR and MDR required detailed agreements covering Intellectual Property (IP), liability, and data governance. Ethically, the project ensured algorithmic transparency, monitored geographic bias, promoted inclusive consent practices, and reinforced equitable access to diagnostic services [58].

Quantitative Analysis The case demonstrates high technological maturity (4.2/5), strong ethical awareness (4.0), solid economic viability (3.8), and moderate legal maturity (3.5). Key tensions include algorithmic traceability versus privacy (tech-law) and efficiency versus equity (economy-ethics). Conversely, tech-economy alignment is strong. Critical success factors include early stakeholder consensus, well-defined IP frameworks, 75% EU co-funding, and unified technical leadership. However, multi-jurisdictional audit complexity required iterative refinement. The overall *Risk-Opportunity Score* is 7.2/10, indicating high potential for successful implementation.

4.2 Edge AI for Chronic Patient Monitoring

DAI in wearable devices has emerged as a key innovation for managing chronic conditions such as diabetes, hypertension, and heart failure. The system analyzed combines smartwatches, glucose sensors, and Electrocardiogram (ECG) devices to enable continuous patient monitoring while preserving privacy through local computation and periodic FL updates [52,2,44].

Application of the Multidimensional Framework The solution uses energy-efficient Edge AI, 5G communication, and TinyML. As demonstrated by Liberti et al. [34], containerized approaches using Kubernetes can significantly improve the deployment flexibility and resource management in such heterogeneous edge environments, addressing key challenges in FL implementation. Economically, it reduces hospitalizations by 30%, employs a pay-per-use AI analysis model, and saves €2000 per patient per year. Legally, challenges include MDR compliance, granular consent, and liability in automated decisions. Ethically, the system raises concerns about patient autonomy, digital divide, and transparency.

Quantitative Analysis Technological maturity scores 4.5/5, followed by economics (4.4), ethics (3.6), and legal (3.2). High tensions exist in tech-ethics and tech-law interactions. Tech-economy alignment is low-conflict. Success drivers include Edge-TinyML readiness, user acceptance, and demonstrable ROI, while regulatory complexity and generational inequity remain barriers. The system scores 8.1/10, suggesting strong success potential.

4.3 Collaborative AI for Genomic Drug Discovery

This case explores a cross-institutional platform inspired by MELLODDY, uniting 15 partners to accelerate drug discovery using federated analysis of genomic, molecular, and clinical data [25,33,24,43].

Application of the Multidimensional Framework The technological axis includes split learning, homomorphic encryption, Quantitative Structure-Activity Relationship (QSAR) pipelines, and multi-omics integration. Economically, it supports public-private benefit-sharing, reduces R&D costs by 40%, and shortens time-to-market by 2–3 years. Legally, the framework manages EU/United States

of America (US)/United Kingdom (UK) compliance, shared patents, and AI liability. Ethically, it addresses global equity, data diversity, and benefit distribution.

Quantitative Analysis The economic axis ranks highest (4.6/5), followed by technology (3.9), ethics (3.3), and law (2.9). Key tensions include IP protection versus collaboration (law-economy) and equity versus profit (ethics-economy). Success depends on encryption expertise and prior MELLODDY validation but is challenged by IP complexity, Western data bias, and competition. The *Risk-Opportunity Score* is 6.8/10, indicating moderate viability.

Methodological Note These case studies are analyzed using the Multidimensional Framework for DAI in Healthcare, which integrates technological, economic, legal, and ethical perspectives. Quantitative assessments are based on evidence-informed metrics derived from real-world implementations, peer-reviewed literature, and industry best practices, enabling predictive and comparative evaluation of implementation success.

5 Discussion

Empirical Relevance and Methodological Triangulation The empirical findings presented by Stile et al. [54] highlight the importance of a mixed-method approach for assessing the socio-organizational impact of adopting BI and AI systems. By integrating structured surveys with focus groups, the study applies methodological triangulation [8,60] to capture both quantifiable changes (e.g., operational efficiency, decision centralization) and subjective dynamics (e.g., cultural resistance, role redefinition). Quantitative data revealed a significant correlation between the adoption of data analytics tools and increased decision-making transparency. However, it also showed a tendency toward informational power centralization among technical actors. While 51.9% of survey participants perceived a "moderately significant" cultural impact, qualitative inputs from focus groups revealed a more nuanced picture, including perceived benefits alongside tensions related to infrastructure and the need for continuous training. The study thus provides a valuable contribution to this research by demonstrating how mixed methods can effectively uncover the multifaceted nature of digital transformation. The empirical validation of themes such as shifts in power structures, evolving competencies, and trust dynamics offers a meaningful point of comparison with our multidimensional framework, reinforcing its operational relevance in real-world healthcare AI implementations.

Organizational Resistance. Our case analysis reveals that organizational resistance operates at three interconnected levels in DAI implementations. At the *individual level*, healthcare professionals' concerns about skill obsolescence create adoption barriers—the FL radiology case showed 60% resistance reduction through early clinician involvement, confirming recent findings on professional

expertise preservation [31]. At the *institutional level*, legacy systems and risk-averse cultures impede transformation; MELLODDY addressed this through phased implementation, aligning with Zhang et al.'s [63] readiness framework. At the *inter-organizational level*, trust deficits between institutions amplify resistance in distributed settings, requiring governance structures that preserve local autonomy while enabling collaboration [39]. Key mitigation strategies include co-design approaches, transparent shared metrics, and reversibility options. These findings highlight that distributed architectures face unique challenges where resistance can cascade across network nodes, requiring simultaneous intervention at all three levels.

6 Conclusions

DAI offers a transformative pathway to reimagine governance in complex healthcare ecosystems. FL and Edge AI enable privacy-preserving collaboration across institutional and regulatory boundaries. However, as our multidimensional framework shows, technological innovation alone is not sufficient. The three case studies reveal that while FL demonstrates strong performance, especially in radiology and chronic care, legal and ethical readiness remains uneven. Key tensions include data sharing, algorithmic accountability, and equitable access, requiring more than technical fixes [9]. DAI also challenges traditional governance, calling for more participatory and adaptive models. New organizational forms, such as consortia and peer-to-peer federations, demand rethinking liability, trust, and control mechanisms [61]. Embedding governance principles from the outset is essential. At the same time, the shift toward patient-centered care requires AI to align not just with efficiency, but also with transparency, fairness, and autonomy. As Topol argues, human empathy and clinical judgment must guide machine intelligence [56]. This study provides a framework for identifying and navigating tensions across four dimensions [2] supporting more informed and responsible AI deployment in healthcare. Future work should examine how AI adoption shapes institutional behavior, focusing on evolving competencies, cultural resistance, and regulatory agility. Longitudinal evaluations will be critical to assess real-world sustainability and equity impacts at scale. The framework's generalizzability beyond European contexts requires validation through: (i) adaptation matrices for non-GDPR jurisdictions, (ii) cultural calibration protocols for ethical metrics, and (iii) economic models accommodating diverse healthcare systems. Supplementary materials detail these extensions.

6.1 Limitations and Future Work

This study faces several methodological and scope limitations:

Methodological Constraints: Our exclusive reliance on secondary sources limits insight into tacit organizational knowledge, informal governance practices,

[2] Technical, economic, legal, and ethical dimension.

and undocumented failures. The document-based scoring methodology, while systematic, cannot capture the full complexity of implementation dynamics or political negotiations that shape real-world deployments.

Geographic and Regulatory Scope: The framework's European regulatory focus, particularly the detailed treatment of Italian implementation, may limit applicability in jurisdictions with different legal traditions or healthcare structures.

Temporal Validity: Given the rapid evolution of both AI technology and regulatory frameworks, our assessments represent a snapshot that requires continuous updating.

Validation Gap: Without primary data from implementers, our framework remains theoretically robust but empirically unvalidated in terms of its predictive power for implementation success.

Future research should prioritize: (1) primary data collection through structured interviews with implementation teams; (2) longitudinal tracking of implementations from inception through maturity; (3) validation in non-European contexts, particularly in emerging economies; (4) development of automated assessment tools using the framework's metrics; and (5) investigation of the framework's predictive validity for implementation outcomes.

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Acronyms

ADR	Alternative Dispute Resolution 14
AI	Artificial Intelligence 1, 2, 3, 4, 5, 6, 7, 9, 12, 13, 14, 15, 16, 20, 21, 22, 23
BI	Business Intelligence 2, 21
CT	Computed Tomography 19
DAI	Distributed Artificial Intelligence 1, 2, 3, 4, 5, 10, 11, 12, 14, 15, 16, 17, 18, 20, 21, 22
DICOM	Digital Imaging and Communications in Medicine 19
DPIA	Data Protection Impact Assessment 13
ECG	Electrocardiogram 20
EHDS	European Health Data Space 5, 13, 17
EU	European Union 2, 10, 13, 14, 20
FDA	Food and Drug Administration 7
FHIR	Fast Healthcare Interoperability Resources 17
FL	Federated Learning 2, 3, 12, 18, 19, 20, 21, 22
GDPR	General Data Protection Regulation 1, 2, 5, 6, 9, 13, 14, 17, 19
HIPAA	Health Insurance Portability and Accountability Act 9, 17
HL7	Health Level Seven International 17
IEC	International Electrotechnical Commission 7
IoT	Internet of Things 11
IP	Intellectual Property 19, 21
ISO	International Organization for Standardization 7
IT	Information Technology 3, 19
MDR	Medical Device Regulation 13, 19, 20
ML	Machine Learning 7
NHS	National Health Service 7
NIHR	National Institute for Health and Care Research 12
QSAR	Quantitative Structure-Activity Relationship 20
ROI	Return on Investment 11, 16, 17, 18, 20
SDGs	Sustainable Development Goals 4
UK	United Kingdom 21
US	United States of America 20
WHO	World Health Organization 6, 10